Introduction to Industrial Management (MEM-112)

Lecture 1

What is Management?

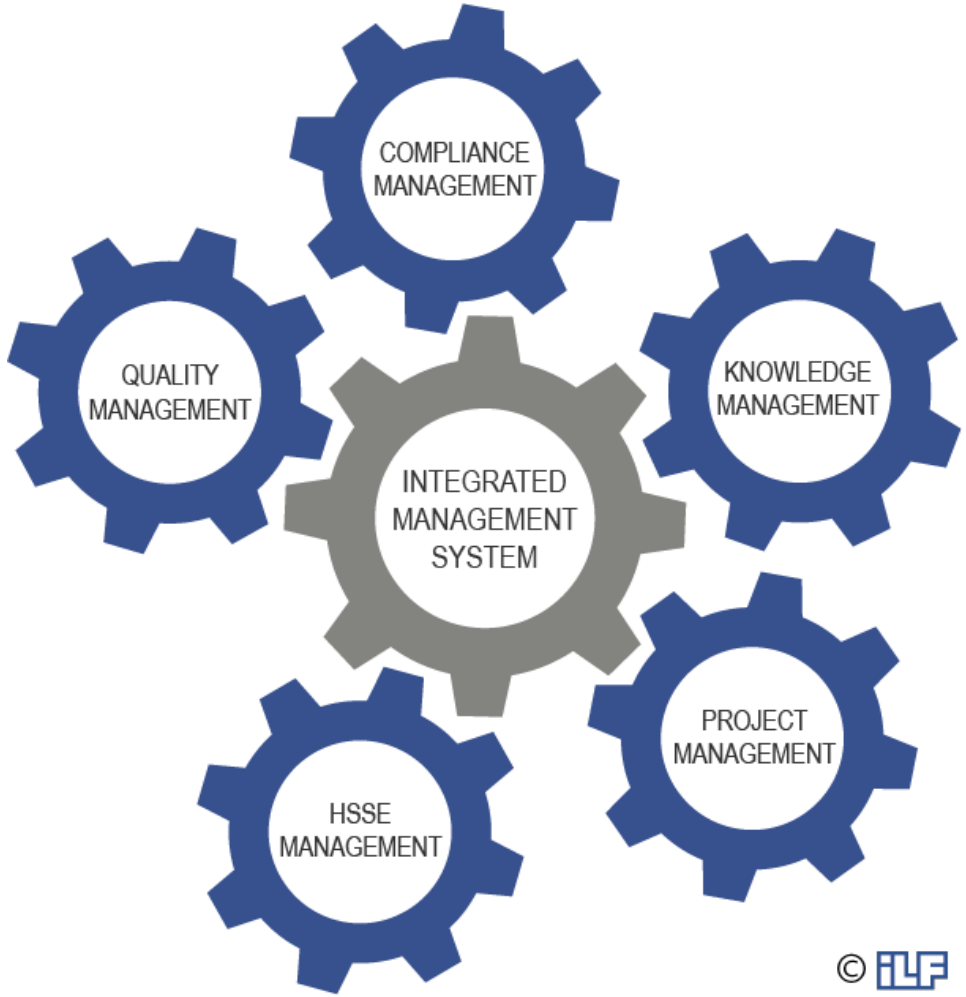
- **Management** is the process of **planning, organizing, leading, and controlling resources** to achieve organizational goals **efficiently and effectively**
- **Examples:**
- Running a factory
- Managing a university department
- Organizing a hospital

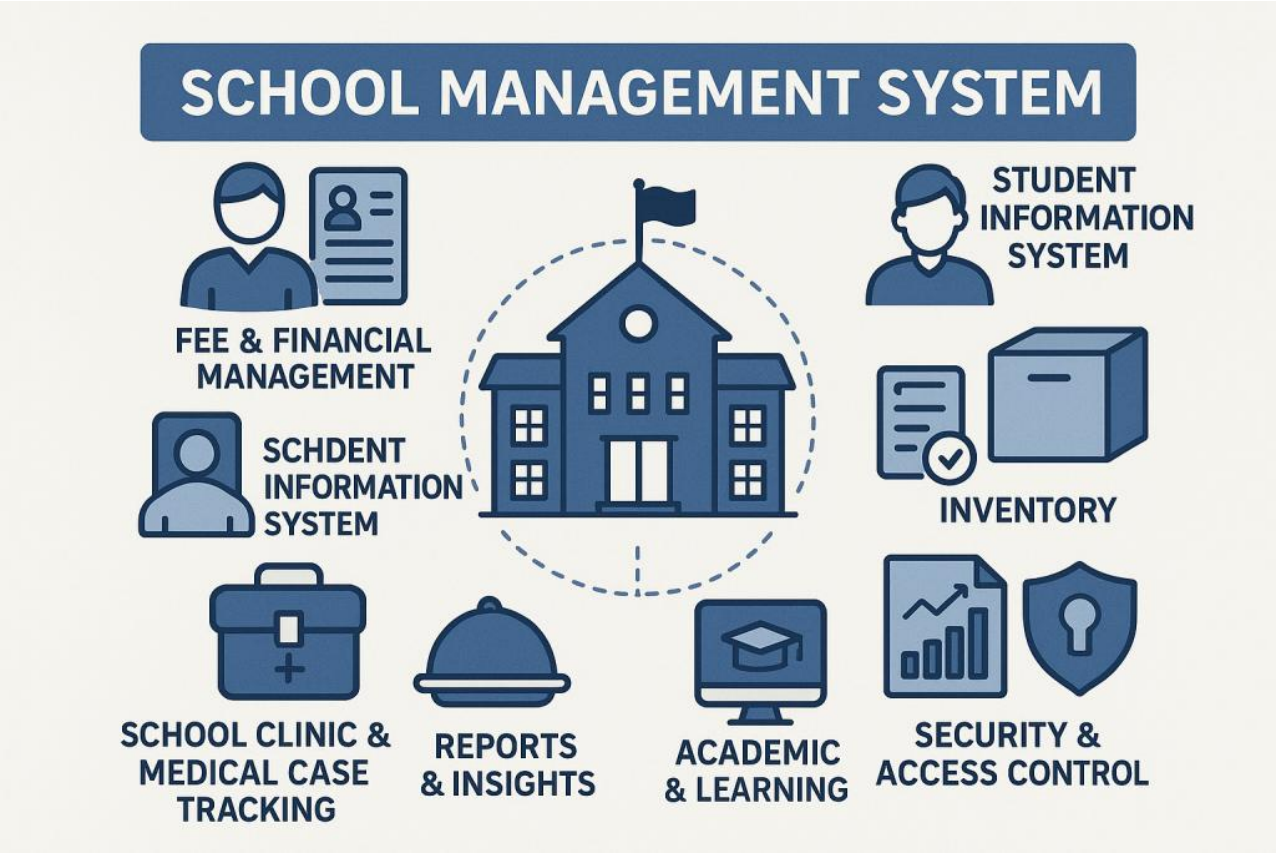
Definition of Management

Henri Fayol:
“Management is to forecast, plan, organize, command, coordinate and control.”

Peter Drucker:
“Management is the art of getting things done through people.”

Management Systems





Industrial Management

INDUSTRIAL MANAGEMENT

- Industrial management is now a branch of engineering which facilitates creation of management system and integrates same with people and their activities to utilize the resources.
- Industrial management is structured approach to manage the operational activities of the organization.

Importance of Industrial Management

Management is important because it:

- Helps achieve organizational goals
- Improves efficiency and productivity
- Reduces wastage of time, money, and materials
- Maintains coordination among departments
- Helps organizations survive competition

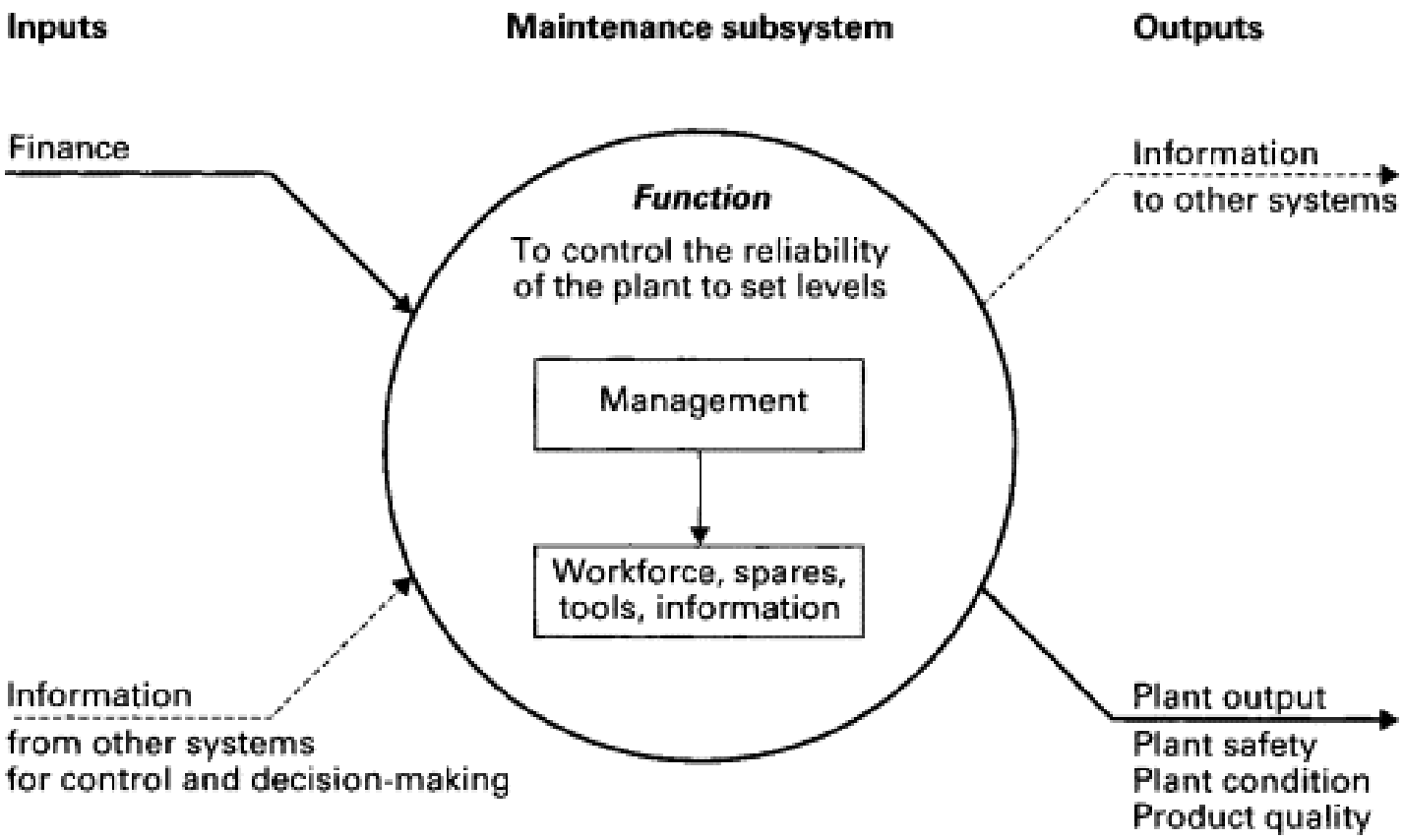
Without management:

- Machines are idle
- Workers are confused
- Raw material is wasted
- Deadlines are missed

Industrial Engineering Techniques

Product Design Efficiency Improvement: Value Engineering	Process Efficiency Improvement	Product/ Process Optimization	Application of Statistics to Improve Efficiency of Processes: Six Sigma
Engineering Economic Analysis	Human Effort Engineering: Motion Analysis	Measurement Work, Cost, Productivity	Productivity Management

Industrial Management



Vision of Management

- A **vision** describes **where the organization wants to go in the future.**
- **Characteristics of a Good Vision:**

❖Long-term

❖Inspiring

❖Clear

❖Future-oriented

- **Industrial Example:**

A power plant's vision may be:

"To provide reliable and clean energy for national development"

Mission of Management

What is Mission?

A **mission** explains **what the organization does, how it does it, and for whom**.

- **Characteristics of a Good Mission:**

❖ Practical

❖ Present-focused

❖ Customer-oriented

❖ Action-based

Example:

- *"To produce high-quality products using modern technology while ensuring customer satisfaction."*

Vision	Mission
Future-oriented	Present-oriented
Dream	Action
Long-term	Short to medium-term
What we want to become	What we do now

Objectives and Goals of Management

Objectives of Management:

- Profitability
- Growth and expansion
- Customer satisfaction
- Employee welfare
- Social responsibility

Example:

- Increase production by 10%
- Reduce cost by improving efficiency
- Improve product quality

Role of Management in Industrial Organizations

Management plays a key role in:

1. Planning production
2. Managing workforce
3. Allocating resources
4. Ensuring quality
5. Meeting deadlines
6. Handling risks

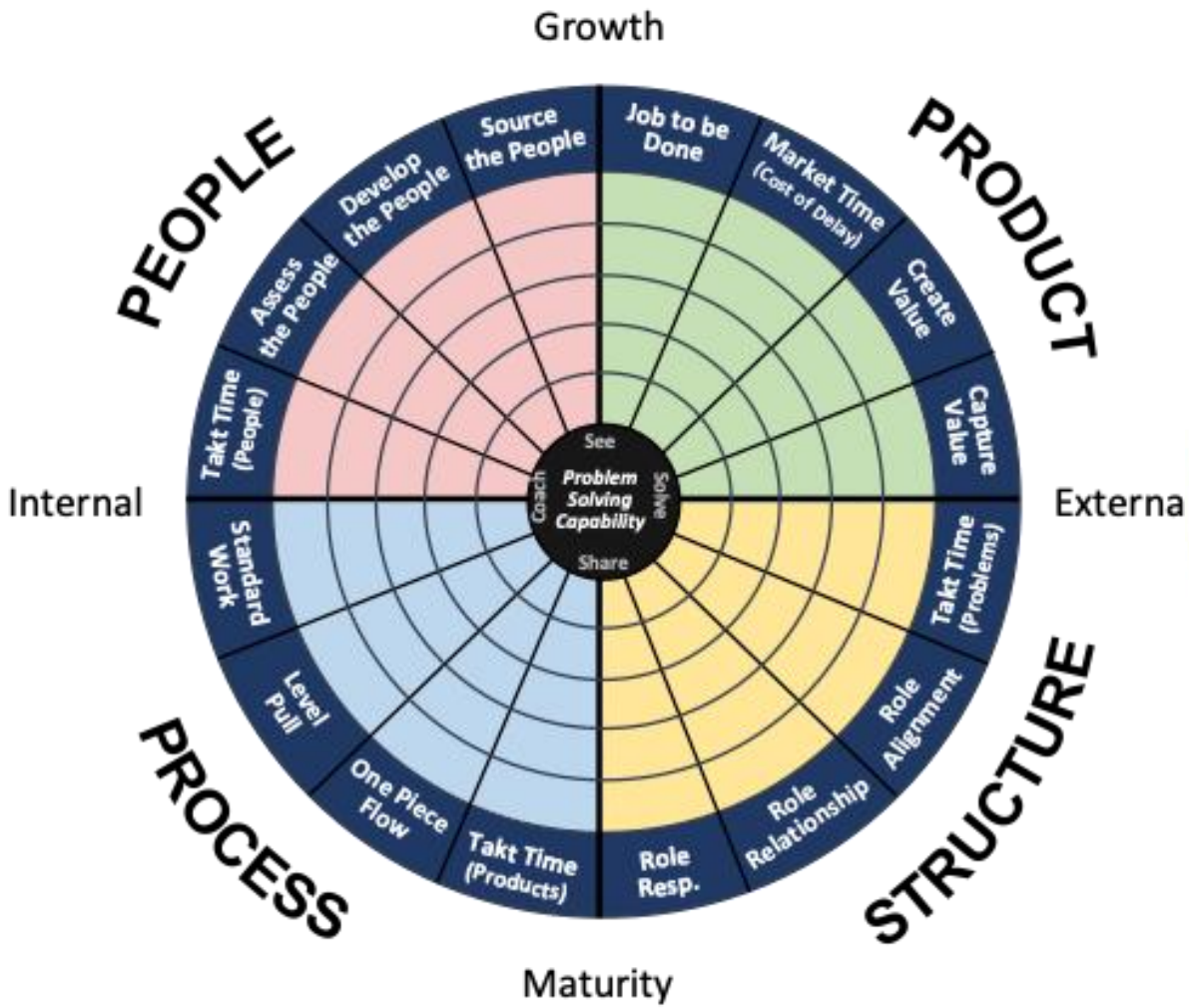
Example:

In a textile industry:

- Management decides production targets
- Supervises workers
- Controls quality
- Manages supply of raw cotton

Management Process

Lecture 2


Growth
PEOPLE
PRODUCT
Internal
Externa
PROCESS
STRUCTURE
Maturity
Source the People
Develop the People
Assess the People
Takt Time (People)
Job to be Done
Market Time (Cost of Delay)
Create Value
Capture Value
Takt Time (Problems)
Role Alignment
Role Relationship
Role Resp.
Takt Time (Products)
One Piece Flow
Level Pull
Standard Work
See
Coach
Problem Solving Capability
Solve
Share

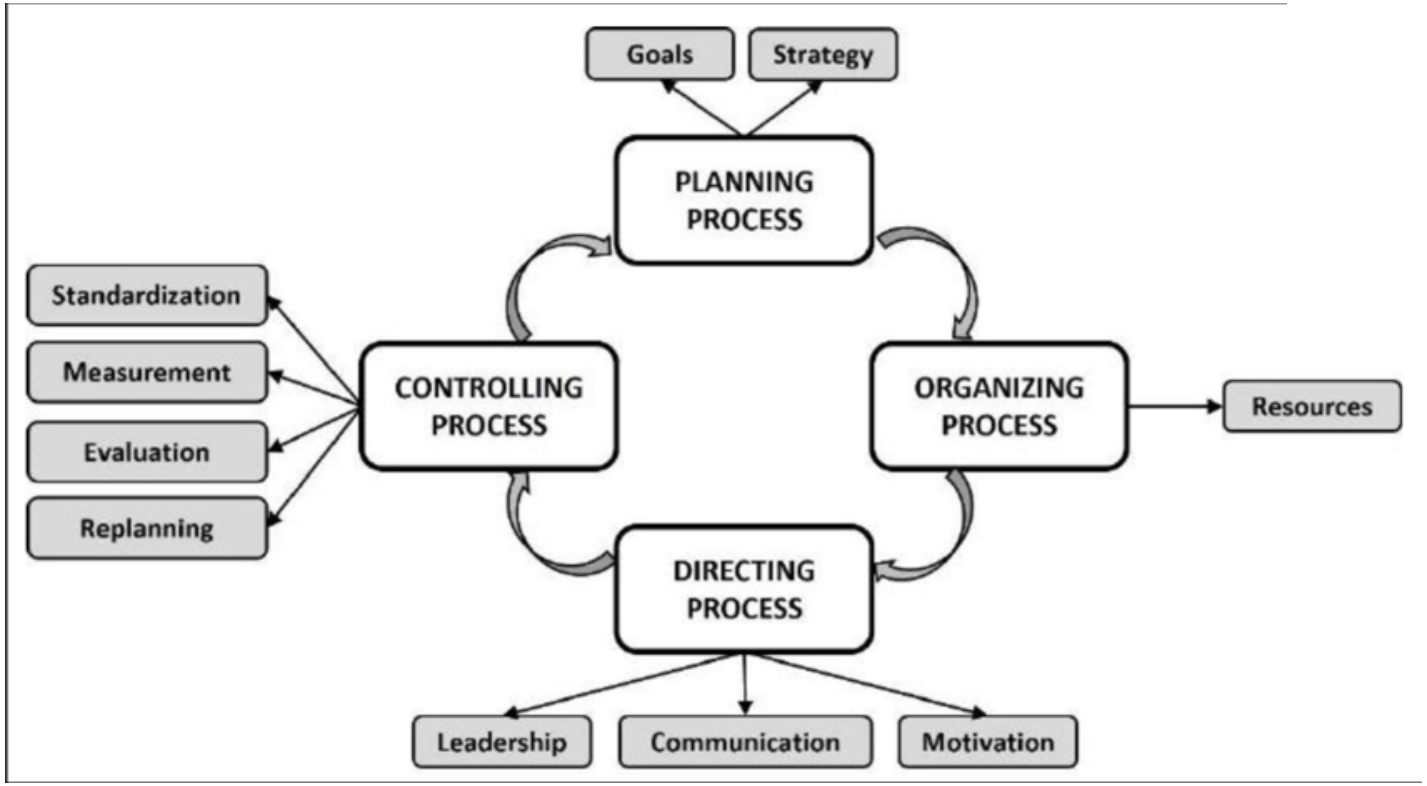
Measure performance
Increase productivity
Optimize operational efficiencies
Connect machines
Reduce loss – increase product margins
MANUFACTURING OPERATIONS MANAGEMENT (MOM) SOLUTION

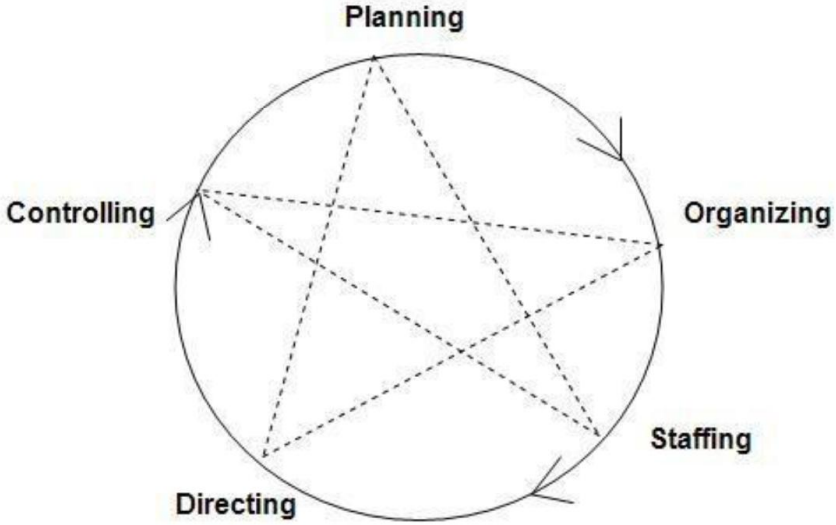
Management Process

The **management process** is a series of activities that managers perform continuously to achieve organizational goals efficiently and effectively.

- **Planning** – Deciding what to do and how to do it
- **Organising** – Arranging resources and assigning responsibilities
- **Staffing:**
- **Directing** – Guiding, motivating, and leading employees
- **Controlling** – Monitoring performance and taking corrective actions

Management Process





Planning

- Decides **what goals to achieve** and **how to achieve them**
- According to KOONTZ, “Planning is deciding in advance - what to do, when to do & how to do. It bridges the gap from where we are & where we want to be”.
- A plan is a future course of actions. It is an exercise in problem solving & decision making
- Types: Strategic (long-term) and Operational (short-term)
  Example: Plan to reduce production costs by 10% in 6 months

Organizing

It is the process of bringing together physical, financial and human resources and developing productive relationship amongst them for achievement of organizational goals.

According to Henry Fayol, “To organize a business is to provide it with everything useful or its functioning i.e. raw material, tools, capital and personnel’s”.

Organizing as a process involves:

- ✓ Identification of activities.
- ✓ Classification of grouping of activities.
- ✓ Assignment of duties.
- ✓ Delegation of authority and creation of responsibility
- ✓ Coordinating authority and responsibility relationships.

Staffing

Staffing is the function of **managing the organization structure and keeping it manned**.
Importance has increased due to:
- ❖Advancement of technology
- ❖Increase in size of business
- ❖Complexity of human behavior

- **Purpose of Staffing**
- To **put the right person in the right job**
  - “Square pegs in square holes, round pegs in round holes”
- **Koontz & O’Donnell:**
  "Managerial function of staffing involves manning the organization structure through proper and effective selection, appraisal & development of personnel to fill the roles designed in the structure."

Components of Staffing

1. **Manpower Planning**
   - Estimating required manpower
   - Choosing the right person
   - Assigning the right place
2. **Recruitment, Selection & Placement**
   - Attracting candidates
   - Selecting best fit
   - Placing them appropriately
3. **Training & Development**
   - Enhancing employee skills
   - Preparing for future roles
4. **Remuneration**
   - Salary, incentives, and benefits
5. **Performance Appraisal**
   - Evaluating employee performance
6. **Promotions & Transfers**
   - Moving employees to higher positions or new roles